

$\pi_{0.5}$ · arXiv 2504.16054 · Physical Intelligence

Empiricist

11+3 · seat 5

H1-H3 / H5 · π_0 VS $\pi_{0.5}$

HL on/off · horizon $H \in \{8, 16, 32, 50\}$

Sep 30 paper club · VLA Architectures

Spine: open-world + hybrid FAST discrete pretrain → FM retained

Semantic subtask → FM chunk (~50-step/~1s @ 50 Hz) · NOT a classical planner · RTC companion

Club spine · one-liner

Open-world VLA via heterogeneous co-training + hierarchical inference:

FAST discrete pretrain → FM retained for continuous control.

Semantic subtask → ~50-step / ~1s FM chunk @ 50 Hz. NOT a classical planner.

RTC companion for async soft-inpaint. Blogs: “FAST replaced FM” = FALSE.

BEHAVIOR'26: $\pi 0.5$ + GR00T baselines · Oct 2 = $\pi 0.5$ -baseline + deltas.

Empiricist hypotheses · H1-H3 / H5

H1

$\pi_{0.5} > \pi_0$

H2

HL on > flat

H3

Co-train morals

H5

Mid horizon

Multi-stage / lang-heavy under match explicit subtask → action raises long-chain WD helps OOD lang; robot diversity ~32 often beats raw H=50 open-loop
Ordinal gap even if SR < paper homes; gap largest on pick-place chains helps OOD scenes (Fig 10-11 direction) BEHAVIOR-like tasks (Comet prior)

Falsifiers: $\pi_0 \geq \pi_{0.5}$ · HL on=off · WD hurts OOD · robot-only=mix · H=8 always best

Experiment matrix · priority order

P0	π_0 VS $\pi_{0.5}$ (OpenPI/Comet)	Success/Q · lang-follow · matched FT
P0	HL on/off same ckpt	Δ on multi-stage tasks
P0	Held-out scene proxy	Progress ID vs OOD
P1	Co-train ablate WD/ME/CE	OOD lang vs manip
P1	Location scaling	Few→many scenes @ fixed N
P1	$H \in \{8,16,32,50\}$	Success vs latency
P2	NFE grid · VI/HL proxy	ms/chunk · HL accuracy
Defer	Full private 280k+80k MM	Out of quiet scope

GPU honesty · budget notes

Workload	GPUs	VRAM	Wall
Eval-only HL / NFE	1 GPU	12-24 GB	hours
Short SFT LoRA 1-2 tasks	1 GPU	24-48 GB	1-3 days
Matched π_0 vs $\pi_{0.5}$ FT	1-2 GPU	24-80 GB	2-6 days
Open-world holdout	1-2 GPU	24-48 GB	days + sim I/O
Knowledge-mix ×4	1-2 GPU	24-48 GB	week-scale if honest
Paper 280k+80k private	multi-node	large	OUT OF SCOPE

Always log (policy, HL, mix, scene, H, Hz, NFE, ms, GPU) next to every success number

π_0 VS $\pi_{0.5}$ · what changed

π_0 (2410.24164)

- FM action expert throughout
- No FAST pre-phase
- No unified HL head in recipe
- Lab / matched scenes focus
- ~10k h cross-embodiment
- H=50 · ~10 Euler · 20-50 Hz
- PaliGemma + ~300M expert

$\pi_{0.5}$ (2504.16054)

- FAST pretrain → FM post-train
- FM RETAINED at inference
- Unified HL subtask → LL chunk
- Open-world new homes KPI
- ~400h MM / ~100 homes + mix
- H=50 @ 50 Hz · 10 FM steps
- Co-train: MM ME CE HL WD VI

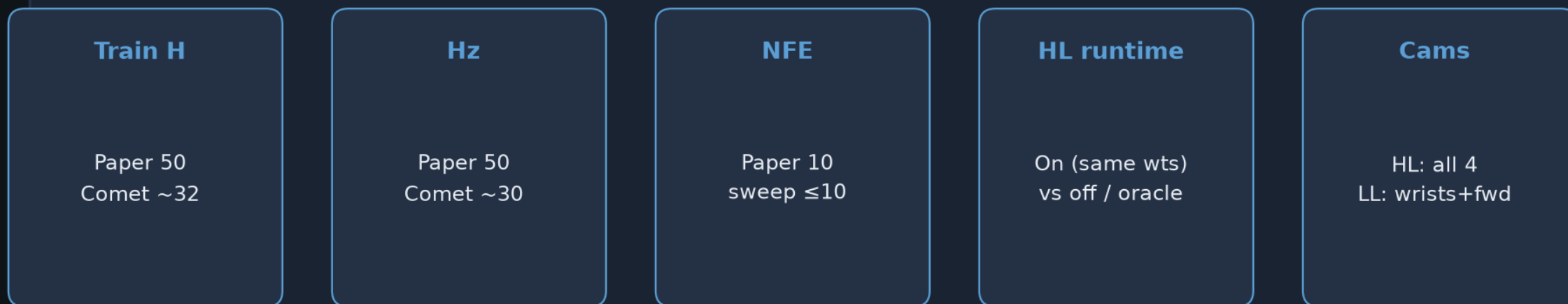
Novelty = mixture curriculum + hierarchy + hybrid recipe + home-scale eval — not new FM math

Hierarchical control loop · semantic subtask → FM chunk

Outer semantic loop · inner continuous control · NOT a classical planner



Control knobs (paper vs challenge priors)



Hybrid recipe · FAST discrete pretrain → FM retained

Bust the myth: FAST does NOT replace FM at deploy

① PRE-TRAIN · 280k steps

Discrete next-token VLA

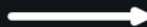
SigLIP 400M + Gemma ~2B

text · boxes · FAST action tokens · $\alpha=0$

FAST action tokens

scalable discrete pretrain signal

Mixture: MM · ME · CE · HL · WD



② POST-TRAIN / INFER · 80k

Joint NTP + flow matching $\alpha=10$

Action expert ~300M · FM

Ada-RMSNorm on τ · random init expert

Inference: AR subtask → 10 FM steps

~50-step / ~1s chunk @ 50 Hz PD

MM+ME+WD+VI · drop CE · FM RETAINED

NOT classical planner · same model for HL + LL · RTC = async soft-inpaint companion (2506.07339)

environments present persistent challenges (e.g., unfamiliar handles on drawers, or cabinets that are physically hard for the robot to open), some behaviors present challenges with partial observability (e.g., the robot arm occluding a spill that should be wiped), and in some cases the high-level sub-task inference is easily distracted (e.g., closing and opening a drawer multiple times while putting away items). Addressing these challenges with better co-training, transfer, and larger datasets is a promising direction for future work. Other future work directions could address the technical constraints of our method. While $\pi_{0.5}$ can perform a variety of behaviors to clean up kitchens and bedrooms, it processes relatively simple prompts. The complexity of the prompts that the model can accommodate is determined by the training data, and more complex preferences and instructions could be incorporated by producing more intricate and diverse annotations, either with human labelers or synthetically. The model also uses a relatively modest context, and incorporating richer context and memory could make the model significantly more capable in settings with more partial observability, such as tasks that require navigating between different rooms or remembering where objects are stored. More broadly, $\pi_{0.5}$ explores a particular combination of heterogeneous data sources, but the specific sources of data can be explored even more broadly. For instance, the ability of our system to learn from verbal instructions provides a powerful new supervision modality, and future work could explore this and other ways that people can provide robots with additional contextual knowledge. We hope that our work will serve as a foundation for a new generation of VLAs that exhibit broad generalization to diverse real-world

Fig.13 · High-level inference ablations

- scale manipulation platform for scalable and intelligent embodied systems. *arXiv preprint arXiv:2503.06669*, 2025.
- [2] Michael Ahn, Anthony Brohan, Noah Brown, Yevgen Chebotar, Omar Cortes, Byron David, Chelsea Finn, Chuyuan Fu, Keerthana Gopalakrishnan, Karol Hausman, Alex Herzog, Daniel Ho, Jasmine Hsu, Julian Ibarz, Brian Ichter, Alex Irpan, Eric Jang, Rosario Jauregui Ruano, Kyle Jeffrey, Sally Jesmonth, Nikhil Joshi, Ryan Julian, Dmitry Kalashnikov, Yuheng Kuang, Kuang-Huei Lee, Sergey Levine, Yao Lu, Linda Luu, Carolina Parada, Peter Pastor, Jornell Quiambao, Kanishka Rao, Jarek Rettinghouse, Diego Reyes, Pierre Sermanet, Nicolas Sievers, Clayton Tan, Alexander Toshev, Vincent Vanhoucke, Fei Xia, Ted Xiao, Peng Xu, Sichun Xu, Mengyuan Yan, and Andy Zeng. Do as i can and not as i say: Grounding language in robotic affordances. In *arXiv preprint arXiv:2204.01691*, 2022.
- [3] Suneel Belkhale and Dorsa Sadigh. Minivla: A better vla with a smaller footprint, 2024. URL <https://github.com/Stanford-ILIAD/opensvla-mini>.
- [4] Suneel Belkhale, Tianli Ding, Ted Xiao, Pierre Sermanet, Quon Vuong, Jonathan Tompson, Yevgen Chebotar, Debiddatta Dwibedi, and Dorsa Sadigh. Rt-h: Action hierarchies using language, 2024. URL <https://arxiv.org/abs/2403.01823>.
- [5] Lucas Beyer, Andreas Steiner, André Susano Pinto, Alexander Kolesnikov, Xiao Wang, Daniel Salz, Maxim Neumann, Ibrahim Alabdulmohsin, Michael Tschannen, Emanuele Bugliarello, et al. Paligemma: A versatile 3b vlm for transfer. *arXiv preprint arXiv:2407.07726*, 2024.
- [6] Homanga Bharadhwaj, Jay Vakil, Mohit Sharma, Abhinav Gupta, Shubham Tulsiani, and Vikash Kumar

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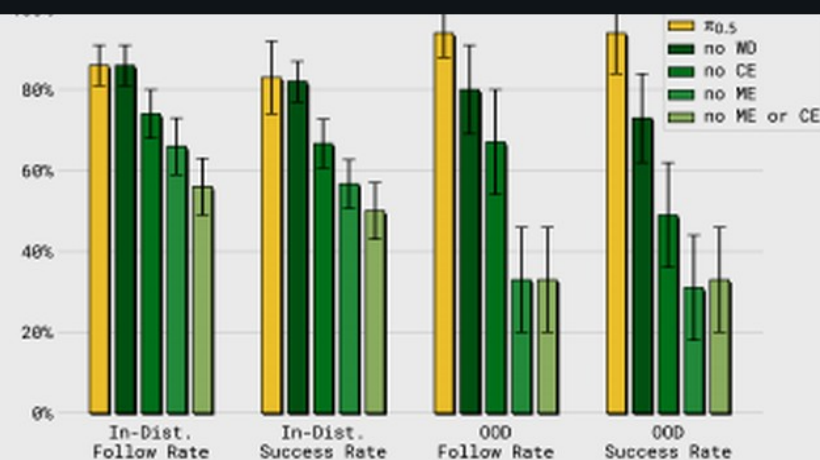


Fig. 11: **Training recipe ablations, language following.** Evaluating language following with in-distribution and out-of-distribution objects after training on different numbers of locations. Including web data (WD) is important for out-of-distribution (OOD) performance in particular. Cross-embodiment (CE) and diverse environment (ME) data both have a large impact on in-distribution and out-of-distribution performance.

difference in performance with the **no WD** ablation is not statistically significant in this experiment, though we show later that web data has a large impact on language following (below) and high-level subtask inference (Section V-E).

The results of the language following experiment, shown in Figure 11, show a similar trend as Figure 10 — excluding ME or/and CE data leads to a significant degradation in performance. What differs now is that removing web data (**no WD**) causes significantly worse performance on out-of-distribution (OOD) objects — we conjecture that training with web data, which contains very broad knowledge of physical objects, allows the model to understand and follow language commands involving *unseen* object categories.

D. How does $\pi_{0.5}$ compare to other VLAs?

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Co-training mixture · MM ME CE HL WD VI

97.6% of phase-1 examples are NOT target mobile household teleop

MM

~400h / ~100 homes

Target mobile manip

ME

Static multi-env

Cross-emb + diversity

CE

Lab + OXE

Task diversity

HL

Subtask (+boxes)

Hierarchy supervision

WD

Web VQA/caption

OOD semantics

VI

~11% of HL-MM

Verbal coaching

Empiricist · close

P0: π_0 vs $\pi_{0.5}$ × HL on/off × held-out scene. Budget-honest.

Quote success only with (policy, HL, mix, scene, H, Hz, NFE, ms, GPU)

Q&A · Sep 30 · $\pi_{0.5}$ / 2504.16054

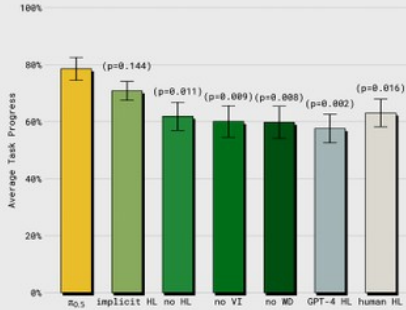


Fig. 13: **Evaluation of the high-level inference process.** While the full $\pi_{0.5}$ model with high-level and low-level inference attains the best results, using only low-level inference (“implicit HL”) with the full $\pi_{0.5}$ model also benefits from the inclusion of high-level subtask examples in training. In contrast, excluding verbal instructions (no VI) or web data (no WD) leads to a significant degradation in performance, and zero-shot prompting a large API-based model (GPT-4) performs worse.

- 3) **no VI**: an ablation of $\pi_{0.5}$ that excludes the verbal instruction (VI) data.
- 4) **implicit HL**: no high-level inference at runtime but includes high-level data in training, which may teach the model about subtasks implicitly.
- 5) **no HL**: no high-level inference, and no high-level data in training at all.
- 6) **GPT-4**: use GPT-4 as the high-level policy, evaluating the importance of training the high-level policy on robot data. To align the model with our domain, we prompt GPT-4 with a description of the task and a list of the most used labels to choose from.
- 7) **human HL**: use an expert human as an “oracle” high-level policy, to provide an upper bound on performance.

The results of these experiments are shown in Figure 13. The full $\pi_{0.5}$ model performs the best, and outperforms even the **human HL** “oracle” baseline. Perhaps surprisingly, the second best model is the **implicit HL** ablation, which does *not* perform any high-level inference, but includes the full data mixture, i.e. also subtask prediction, in training. This strongly suggests the importance of the co-training recipe used by our model: while there is a benefit to explicitly infer high-level subtasks, a significant portion of that benefit is already obtained simply by including subtask *prediction* data in the training mixture. The **no HL** ablation, excluding HL task even in training, performs significantly worse. The results also show that the relatively small verbal instruction dataset, which only constitutes about 11% of the high-level mobile manipulation examples, is critical to strong performance as the **no VI** ablation is significantly weaker. The **no WD** ablation is also significantly worse, indicating that much of the benefit of web data (perhaps unsurprisingly) lies in improving the

high-level policy. Finally, the zero-shot **GPT-4** ablation attains the worst performance, indicating the importance of adapting VLMs with robot data. We provide a detailed breakdown of performance on each task in Appendix D, Figure 17.

VI. DISCUSSION AND FUTURE WORK

We described $\pi_{0.5}$, a co-trained model that builds on the π_0 VLA to integrate a variety of data sources and enable generalization to new environments. The $\pi_{0.5}$ VLA can control mobile manipulators to perform tasks in homes that were never seen in the training data, cleaning kitchens and bedrooms, making beds, hanging towels, and performing other multi-stage and dexterous behaviors. $\pi_{0.5}$ is trained on about 400 hours of mobile manipulation data, but includes a much larger amount of data from other robots, including non-mobile manipulators in diverse environments and data collected under laboratory conditions. It is also co-trained jointly with data from the web, as well as high-level prediction data for outputting language commands based on robot observations. The generalization capabilities of $\pi_{0.5}$ demonstrate that this co-training recipe facilitates effective transfer, enabling highly generalizable control of a mobile manipulator with only a medium-sized mobile manipulation dataset.

$\pi_{0.5}$ is not without its limitations. While our VLA exhibits broad generalization, it still makes mistakes. Some environments present persistent challenges (e.g., unfamiliar handles on drawers, or cabinets that are physically hard for the robot to open), some behaviors present challenges with partial observability (e.g., the robot arm occluding a spill that should be wiped), and in some cases the high-level subtask inference is easily distracted (e.g., closing and opening a drawer multiple times while putting away items). Addressing these challenges with better co-training, transfer, and larger datasets is a promising direction for future work. Other future work directions could address the technical constraints of our method. While $\pi_{0.5}$ can perform a variety of behaviors to clean up kitchens and bedrooms, it processes relatively simple prompts. The complexity of the prompts that the model can accommodate is determined by the training data, and more complex preferences and instructions could be incorporated by producing more intricate and diverse annotations, either with human labelers or synthetically. The model also uses a relatively modest context, and incorporating richer context and memory could make the model significantly more capable in settings with more partial observability, such as tasks that require navigating between different rooms or remembering where objects are stored. More broadly, $\pi_{0.5}$ explores a particular combination of heterogeneous data sources, but the specific sources of data can be explored even more broadly. For instance, the ability of our system to learn from verbal instructions provides a powerful new supervision modality, and future work could explore this and other ways that people can provide robots with additional contextual knowledge. We hope that our work will serve as a foundation for a new generation of VLAs that exhibit broad generalization to diverse real-world environments.